



# PRODUCT SPECIFICATION

## 产品规范

适用于 For  
K902 GNSS 模块  
K902 GNSS Module



### INTRODUCTION

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## REVISION HISTORY / 修订历史

| Revision/版本 | Modification/更改                                             | Date/日期    |
|-------------|-------------------------------------------------------------|------------|
| 1.0         | New Release. / 新发                                           | 2024/09/18 |
| 1.1         | Add an Identification of PIN 1/ 增加一脚标识                      | 2024/12/05 |
| 1.2         | Add Usage Requirements of Pin SYS_RSTN / 增加 SYS_RSTN 引脚使用要求 | 2024/12/18 |
| 1.3         | Add Transformer Connection Mode / 增加变压器连接方式                 | 2025/2/19  |
| 1.4         | Modify the Protect Image / 修改产品图片                           | 2025/3/3   |

## DIRETORY / 目录

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# 1. Introduction / 简介

K902 is a small-size high-precision positioning module independently developed by SinoGNSS at full - system and full-frequency. It tracks constellations including BDS-2, BDS-3, GPS, GLONASS, Galileo and QZSS, also with onboard inertial navigation device, supporting integrated navigation algorithm. The K902 GNSS module is mainly used for surveying and mapping, monitoring, outdoor robotics and so on.

K902模块是司南导航自主研发的全系统全频点的小尺寸高精度定位模块。支持BDS-2、BDS-3、GPS、GLONASS、Galileo和QZSS等卫星导航系统的信号跟踪，板载惯导器件，支持组合导航算法，适用于测量测绘、监测、室外机器人等。

## 1.1. Product Characteristics / 产品特性

Table 1. Product Characteristics / 产品特性

| Characteristics             | K902                                                             |                                         |
|-----------------------------|------------------------------------------------------------------|-----------------------------------------|
| Signals<br>信号               | Positioning<br>定位                                                | GPS: L1C/A, L2P, L2C, L5, L1C           |
|                             |                                                                  | BDS-2: B1I, B2I, B3I                    |
|                             |                                                                  | BDS-3: B1I, B1C, B2a, B2b, B3I          |
|                             |                                                                  | GLONASS: G1, G2, G3*                    |
|                             |                                                                  | Galileo: E1, E5b, E5a, E5 AltBoC*, E6c* |
|                             |                                                                  | QZSS: L1C/A, L2C, L5, L1C               |
|                             |                                                                  | SBAS: L1C/A, L5                         |
|                             |                                                                  | IRNSS*: L5                              |
|                             | Items with * will be adjusted with the version.<br>带*项会随同版本进行调整。 |                                         |
| Time to First Fix<br>首次定位时间 | Cold Start<br>冷启动                                                | 20s                                     |
|                             | Hot Start (with RTC)                                             | 10s                                     |

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|                                |                                              |                                                |
|--------------------------------|----------------------------------------------|------------------------------------------------|
|                                | 热启动（使用RTC）                                   |                                                |
| Signal Capture<br>信号捕获         | Reacquisition<br>失锁重捕                        | 1s                                             |
|                                | Signals Tracking Sensitivity<br>信号跟踪灵敏度      | -155dBm                                        |
|                                | Signal Capture Sensitivity<br>信号捕获灵敏度        | -140dBm                                        |
| Measurement Precision<br>测量准确度 | Pseudo-range Accuracy<br>伪距精度                | 10cm                                           |
|                                | Carrier Phase Accuracy<br>载波相位精度             | 1mm                                            |
| Accuracy<br>精度                 | PPS（RMS）<br>授时精度                             | 20ns                                           |
|                                | SPP Accuracy<br>标准单点定位精度                     | 平面（Horizontality）：1.5m<br>高程（Verticality）：3m   |
|                                | Velocity Accuracy<br>测速精度                    | 0.02m/s                                        |
|                                | Assisted dead reckoning（error）<br>辅助航位推算（误差） | 3%d（d为行驶距离）                                    |
| PPP                            | PPP Convergence Time<br>PPP收敛时间              | 20min                                          |
|                                | PPP Accuracy<br>PPP精度                        | 平面（Horizontality）：0.1m<br>高程（Verticality）：0.2m |

|                          |                                                                                                                                                                                                                          |                                                                                                                                                   |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| RTK                      | RTK Initialization Time<br>RTK初始化时间                                                                                                                                                                                      | 5s (D<10km)                                                                                                                                       |
|                          | Initialization Reliability<br>初始化置信度                                                                                                                                                                                     | 99.9%                                                                                                                                             |
|                          | RTK Accuracy<br>RTK精度                                                                                                                                                                                                    | 平面 (Horizontality) : $(8+10^{-6} \times D)$ mm<br>高程 (Verticality) : $(15+10^{-6} \times D)$ mm<br>D为基线长度(单位: mm)<br>D-Baseline length (Unit: mm) |
| Anti-interference<br>抗干扰 | It can suppress the potential narrowband and single tone radio interference signals in the GNSS signal frequency band, and the interference to signal ratio can reach 60dB.<br>具备抑制GNSS信号频带内潜在的窄带和单音无线电干扰信号, 干信比可达 60dB。 |                                                                                                                                                   |
| IMU                      | Gyroscope<br>陀螺仪                                                                                                                                                                                                         | Measurement ranges<br>量程: $\pm 125^{\circ}/s$                                                                                                     |
|                          |                                                                                                                                                                                                                          | Zero-biased repeatability<br>零偏重复性: $0.5^{\circ}/s$                                                                                               |
|                          |                                                                                                                                                                                                                          | Zero-biased stability<br>零偏稳定性: $5^{\circ}/h$                                                                                                     |
|                          |                                                                                                                                                                                                                          | Angular random walk<br>角度随机游走: $0.12^{\circ}/\sqrt{h}$                                                                                            |
|                          | Accelerometer<br>加速度计                                                                                                                                                                                                    | Measurement ranges<br>量程: $\pm 2g$                                                                                                                |
|                          |                                                                                                                                                                                                                          | Zero-biased repeatability<br>零偏重复性: 20mg                                                                                                          |
|                          |                                                                                                                                                                                                                          | Bias instability                                                                                                                                  |

|                        |                                |                                                                                                                           |
|------------------------|--------------------------------|---------------------------------------------------------------------------------------------------------------------------|
|                        |                                | 零偏稳定性: 50ug                                                                                                               |
|                        |                                | Velocity random walk<br>速度随机游走: 0.07m/s/sqrt(h)                                                                           |
| Data Rates<br>数据速率     | Position<br>单点定位               | Max 50Hz (The default is 5Hz, and the maximum option is 50Hz / 默认 5Hz, 最高可选配至 50Hz)                                       |
|                        | RTK<br>实时载波相位差分                | Max 50Hz (The default is 5Hz, and the maximum option is 50Hz / 默认 5Hz, 最高可选配至 50Hz)                                       |
|                        | IMU                            | Max 100Hz (The default is 5Hz, and the maximum option is 100Hz / 默认 5Hz, 最高可选配至 100Hz)                                    |
| Data Formats<br>输出数据格式 | NMEA-0183                      | GPGLL, GPGGA, GPRMC, GPVTG, GPZDA etc.                                                                                    |
|                        | ComNav Binary (CNB)<br>司南二进制格式 | ComNav Self-Defined Binary<br>司南自定义二进制                                                                                    |
|                        | CMR(GPS)                       | CMROBS, CMRREF                                                                                                            |
|                        | RTCM2.X                        | RTCM1, RTCM3, RTCM9, RTCM1819, RTCM31, RTCM41, RTCM42                                                                     |
|                        | RTCM3.X                        | 1004 ~ 1008, 1012, 1019, 1020, 1033, 1042, 1045/1046, 1230, 4078<br>MSM3~MSM7: 1073~1077, 1083~1087, 1123~1127, 1093~1097 |
| Electrical<br>电气特性     | Voltage<br>供电电压                | 3.3 V ± 5 %                                                                                                               |
|                        | Power Consumption<br>功耗        | 0.4 W (Anti-interference off, 未开启抗干扰)<br>抗干扰功能开启, 功耗约增加 0.1W<br>Set anti-interference on consumes more about 0.1W         |

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|                                   |                                      |                                                                                                                                                                                                                |
|-----------------------------------|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Environmental</b><br>环境要求      | <b>Operating Temperature</b><br>工作温度 | -40°C~+85°C<br><br>注意：模组散热焊盘务必有效接地，有且散热面积不小于模组面积。<br><br>Attention: The module's heat dissipation pad must be effectively grounded, and the heat dissipation area must be no less than the area of the module. |
|                                   | <b>Storage Temperature</b><br>储存温度   | -55°C~+125°C                                                                                                                                                                                                   |
| <b>Antenna Interface</b><br>天线接口  | <b>Impedance Matching</b><br>阻抗匹配    | 50Ω                                                                                                                                                                                                            |
|                                   | <b>Antenna Gain</b><br>天线增益          | 20~35dB                                                                                                                                                                                                        |
| <b>Hardware Interface</b><br>硬件接口 |                                      | UART×2, UART3×1/CAN×1 (二选一)<br><br>I2C×1, SPI×1, PPS×1, EVENT×1<br><br>ETH×1 (ETH对应的引脚为 24、25、28、29)                                                                                                           |
| <b>Physical</b><br>物理参数           | <b>Size</b><br>尺寸                    | 17mm×22mm×2.5mm                                                                                                                                                                                                |
|                                   | <b>Weight</b><br>重量                  | 1.9g                                                                                                                                                                                                           |
|                                   | <b>Package</b><br>封装                 | LGA (54PIN)                                                                                                                                                                                                    |



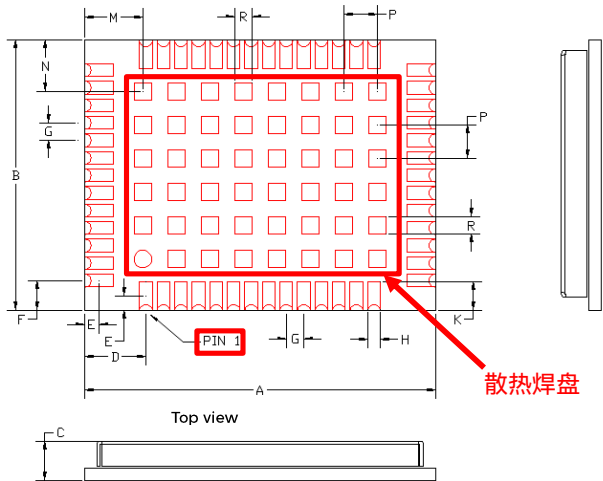
2. Size / 尺寸

In this section, product photo, mechanical drawing and the dimension of K902 is provided for customers' further hardware design and installation.

本节提供了K902的实物图，机械图 and 对应的物理尺寸，便于用户进一步系统硬件设计和安装。



Figure 1. K902 Product Photo / K902 实物图



| 尺寸代码 | 尺寸值 (mm) | 公差 (mm) |
|------|----------|---------|
| A    | 22.00    | ±0.2    |
| B    | 17.00    | ±0.2    |
| C    | 2.50     | ±0.2    |
| D    | 3.85     | -       |
| E    | 0.90     | -       |
| F    | 1.90     | -       |
| G    | 1.10     | -       |
| H    | 0.80     | -       |
| M    | 3.65     | -       |
| N    | 3.25     | -       |
| R    | 1.10     | -       |
| P    | 2.10     | -       |
| K    | 1.80     | -       |

Figure 2. K902 Mechanical Drawing / K902 机械图

### 3. Pin Arrangement and Definition / 针脚标识和定义

K902 is surface-mount OEM Module which integrates 54 Pin.

K902包括54Pin，表贴式模块。

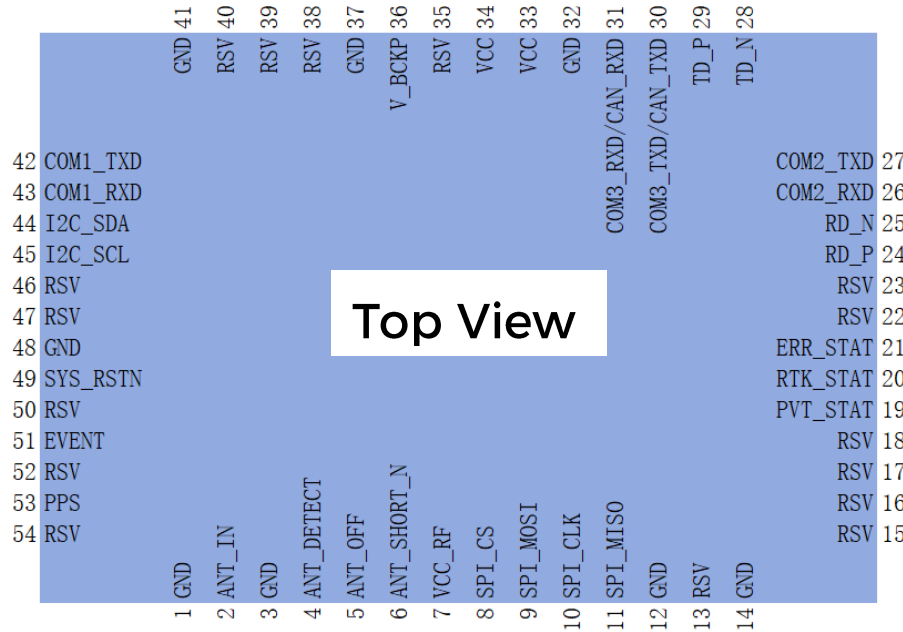


Figure 3. K902 Includes 54-Pin Pad/K902 包括54连接焊盘

Table 2. Pin Definition of K902 54-Pin Pad / K902 54针脚焊盘的针脚定义

| Pin | Signal                    | Type | Description                                      |               |
|-----|---------------------------|------|--------------------------------------------------|---------------|
| 1   | GND                       | GND  | Ground Reference                                 | 参考地           |
| 2   | ANT_IN                    | I    | ANT input                                        | 天线射频输入        |
| 3   | GND                       | GND  | Ground Reference                                 | 参考地           |
| 4   | ANT_DETECT* <sup>1</sup>  | I    | Active antenna detect - default active high      | 天线开路检测-默认高电平  |
| 5   | ANT_OFF* <sup>1</sup>     | O    | External LNA disable - default active high       | 外部低噪放使能-默认高电平 |
| 6   | ANT_SHORT_N* <sup>1</sup> | I    | Active antenna short detect - default active low | 天线短路检测-默认高电平  |
| 7   | VCC_RF* <sup>2</sup>      | O    | Voltage for external LNA                         | 外部低噪放供电       |
| 8   | SPI_CS                    | I/O  | SPI_CS                                           | SPI片选信号       |
| 9   | SPI_MOSI                  | I/O  | SPI_MOSI                                         | SPI主设备数据输出    |
| 10  | SPI_CLK                   | I/O  | SPI_CLK                                          | SPI时钟信号       |

| Pin   | Signal           | Type  | Description                 |                               |
|-------|------------------|-------|-----------------------------|-------------------------------|
| 11    | SPI_MISO         | I/O - | SPI_MISO                    | SPI主设备数据输入                    |
| 12    | GND              | GND   | Ground Reference            | 参考地                           |
| 13    | RSV              | -     | Reserved                    | 保留管脚                          |
| 14    | GND              | GND   | Ground Reference            | 参考地                           |
| 15-18 | RSV              | -     | Reserved                    | 保留管脚                          |
| 19    | PVT_STAT         | O     | Number of search satellites | 搜星数量指示                        |
| 20    | RTK_STAT         | O     | RTK Data Indicator          | 在接收或发送差分信号时会输出高电平脉冲，其他状态输出低电平 |
| 21    | ERR_STAT         | O     | Failure Indication          | 故障指示                          |
| 22-23 | RSV              | -     | Reserved                    | 保留管脚                          |
| 24    | RD_P             | I     | Receive Data+               | 接收数据正                         |
| 25    | RD_N             | I     | Receive Data-               | 接收数据负                         |
| 26    | COM2_RXD         | I     | UART2 receive               | 串口2接收                         |
| 27    | COM2_TXD         | O     | UART2 transmit              | 串口2发送                         |
| 28    | TD_N             | O     | Transmit Data -             | 发送数据负                         |
| 29    | TD_P             | O     | Transmit Data +             | 发送数据正                         |
| 30    | COM3_TXD/CAN_TXD | O     | UART3/CAN transmit          | 串口3或CAN发送（默认为串口3）             |
| 31    | COM3_RXD/CAN_RXD | I     | UART3/CAN receive           | 串口3或CAN接收（默认为串口3）             |
| 32    | GND              | GND   | Ground Reference            | 参考地                           |
| 33~34 | VCC              | PWR   | Voltage supply              | 电源输入                          |
| 35    | RSV              | -     | Reserved                    | 保留管脚                          |
| 36    | V_BCKP           | PWR   | Backup supply voltage       | RTC电源输入                       |
| 37    | GND              | GND   | Ground Reference            | 参考地                           |
| 38-40 | RSV              | -     | Reserved                    | 保留管脚                          |
| 41    | GND              | GND   | Ground Reference            | 参考地                           |
| 42    | COM1_TXD         | O     | UART1 transmit              | 串口1发送                         |
| 43    | COM1_RXD         | I     | UART1 receive               | 串口1接收                         |
| 44    | I2C_SDA          | I/O   | Data Line                   | I2C数据                         |
| 45    | I2C_SCL          | I/O   | Clock Line                  | I2C时钟                         |
| 46-47 | RSV              | -     | Reserved                    | 保留管脚                          |
| 48    | GND              | GND   | Ground Reference            | 参考地                           |
| 49    | SYS_RSTN         | I     | System Reset Negative       | 系统复位，低电平有效                    |
| 50    | RSV              | -     | Reserved                    | 保留管脚                          |
| 51    | EVENT            | I     | External Interrupt Pin      | 外部中断输入                        |
| 52    | RSV              | -     | Reserved                    | 保留管脚                          |
| 53    | PPS              | O     | Pulse Per Second            | 秒脉冲                           |
| 54    | RSV              | -     | Reserved                    | 保留管脚                          |
| Note: |                  |       |                             |                               |

| Pin                                                                                                                                                                                                                                                                                                                                                                        | Signal | Type | Description |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|------|-------------|
| <p>*1. It is not supported temporarily;</p> <p>*2. It is not recommended to use VCC_RF as ANT_BIAS to feed power to the antenna (due to the limited module size, VCC_RF is not equipped with lightning protection, surge protection, and overcurrent protection).</p> <p>说明:</p> <p>*1、暂不支持;</p> <p>*2、VCC_RF 不建议作为 ANT_BIAS 给天线馈电 (因受限于模块体积, VCC_RF 未做防雷击、防浪涌、过流保护处理)</p> |        |      |             |

### 3.1. Remarks / 说明

#### 1. Electrical Characteristics / 电气特性

COM1/2/3(TX&RX), SPI, ANT(DETECT&OFF&SHORT\_N), I2C, SYS\_RSTN, PPS, EVENT, ERR\_STAT, PVT\_STAT and RTK\_STAT are LVCMOS 3.3V. All these signals are compatible with LVCMOS / LVTTTL 3.3V.

COM1/2/3 (TX&RX), SPI, ANT (DETECT&OFF&SHORT\_N), I2C, SYS\_RSTN, PPS, EVENT, ERR\_STAT, PVT\_STAT 和 RTK\_STAT 为 LVCMOS 3.3V 电平, 所有这些信号均兼容 LVCMOS / LVTTTL 3.3V。

Table 3. LVCMOS 3.3V Electrical Standard / LVCMOS 3.3V电气标准

| Symbols 符号      | Description 描述                       | Min 最小 | Max 最大 |
|-----------------|--------------------------------------|--------|--------|
| V <sub>IH</sub> | Input high voltage<br>输入高电压          | 2.0V   | 3.6V   |
| V <sub>IL</sub> | Input low voltage<br>输入低电压           | -0.3V  | 0.8V   |
| V <sub>OH</sub> | High-level output voltage<br>高电平输出电压 | 2.9V   | --     |
| V <sub>OL</sub> | Low-level output voltage<br>低电平输出电压  | --     | 0.4V   |
| I <sub>OH</sub> | Sourcing current<br>拉电流              |        | 8mA    |
| I <sub>OL</sub> | Sinking current<br>灌电流               |        | 8mA    |

Table 4. LVTTTL 3.3V Electrical Standard / LVTTTL 3.3V电气标准

| Symbols 符号      | Description 描述                       | Min 最小   | Max 最大   |
|-----------------|--------------------------------------|----------|----------|
| V <sub>IH</sub> | Input high voltage<br>输入高电压          | 2.0V     | VCC+0.3V |
| V <sub>IL</sub> | Input low voltage<br>输入低电压           | -0.3V    | 0.8V     |
| V <sub>OH</sub> | High-level output voltage<br>高电平输出电压 | VCC-0.4V | --       |

| Symbols 符号 | Description 描述                      | Min 最小 | Max 最大 |
|------------|-------------------------------------|--------|--------|
| $V_{OL}$   | Low-level output voltage<br>低电平输出电压 | --     | 0.41V  |
| $I_{OH}$   | Sourcing current<br>拉电流             |        | 8mA    |
| $I_{OL}$   | Sinking current<br>灌电流              |        | 8mA    |

## 2. Can withstand Voltage Range / 能承受的电压范围

The signal with the maximum voltage range from -0.3V to 3.6V is as follows: COM1/2/3(TX&RX), SPI, I2C, ANT(DETECT&OFF&SHORT\_N), ERR\_STAT, PVT\_STAT, RTK\_STAT, SYS\_RSTN, PPS, EVENT

所能承受电压的最大值范围是 -0.3V~3.6V 的信号如下：COM1 / 2 / 3 (TX&RX)，SPI，I2C，ANT(DETECT&OFF&SHORT\_N)，ERR\_STAT，PVT\_STAT，RTK\_STAT，SYS\_RSTN，PPS，EVENT

## 3. Supply Voltage / 供电电压

VCC is main power supply, and voltage range is 3.3V. The requirement for voltage ripple and spike is less than 50mV. The voltage range of V\_BCKP is 1.8V to 3.6V, and the requirements for voltage ripple and spike are less than 30mV.

VCC主供电电源，电压范围：3.3V（直流）。电压纹波和尖峰脉冲要求小于50mV。V\_BCKP，电压1.8V~3.6V，电压纹波和尖峰脉冲要求小于30mV。

## 4. Power On and Off the module/ 模块上电与下电

VCC/V\_BCKP

- VCC/V\_BCKP power-on, the initial level is lower than 0.4V, and has a good monotonicity, and the upstroke and ringing are guaranteed within the range of 5% VCC/V\_BCKP;
- VCC/V\_BCKP power-on waveform, rise time from 10% to 90% should be in the range from 100us to 1ms.
- Power-on interval, after the VCC/V\_BCKP is powered off less than 0.4V, before the next power-on, the interval must be greater than 500ms.

VCC/V\_BCKP

- VCC/V\_BCKP 上电，起始电平低于 0.4V，具有良好的单调性，下冲与振铃保障在 5% VCC/V\_BCKP 范围内。
- VCC/V\_BCKP 上电波形，从 10%到 90%的上升时间需在 100us~1ms 范围内。
- 上电时间间隔，VCC/V\_BCKP 下电低于 0.4V 后，到下一次开始上电，时间间隔需大于 500ms。

## 5. Usage Requirements of Pin SYS\_RSTN / SYS\_RSTN 引脚使用要求

- The module is equipped with an internal power-on reset circuit, eliminating the need for an external hardware reset circuit;

- If choose to control the reset of SYS\_RSTN through GPIO, ensure that VCC has been properly powered up; then use GPIO to reset SYS\_RSTN by pulling it down to ground (longer than 1ms) and then pulling it up to 3.3V.
- 模块内置上电复位电路，不需要外接硬件复位电路；
- 若通过 GPIO 控制复位 SYS\_RSTN，务必保证 VCC 已正常供电；然后通过 GPIO 把 SYS\_RSTN 下拉到地复位 (时长大于 1ms)，再上拉至 3.3V。

## 6. RTK\_STAT

RTK\_STAT signal pin, receiving or transmitting differential signals will output high level pulse, other states output low level, needing to connect to the external indicator light.

RTK\_STAT信号脚，在接收或发送差分信号时会输出高电平脉冲，其他状态输出低电平，需要外接指示灯。

## 7. Add Surge Protection / 增加浪涌保护

When the user integrates the module, an external high-gain choke antenna is connected. If a surge protector is not added, it is easy to damage antenna feed circuit. It is recommended that users install a surge protector on the antenna when connecting a high-gain choke antenna.

当用户集成模块时，外接高增益的扼流圈天线，如果不加浪涌保护器，容易造成天线馈电电路损坏。建议用户在外接高增益扼流圈天线时，天线上安装一个浪涌保护器。

## 8. Thermal Design / 热设计

The K902 module contains temperature-sensitive components inside, requiring a stable temperature environment; hence, during PCB layout, the module needs to be positioned away from heat-generating or low-temperature areas. The TCXO (Temperature Compensated X Tal Oscillator) inside the module is highly sensitive to temperature changes, and sudden temperature fluctuations can affect the module's ability to track satellite signals. Consequently, the layout of the module should maintain a certain distance from heat-generating components, cooling fans, or heat dissipation vents.

K902模组内部含温度敏感元件，需确保温度稳定；故PCB布局时，模组需远离发热区或低温区。模组内部的TCXO（温度补偿晶体振荡器）对温度变化反应敏感，温度突变会影响模组对卫星信号的追踪能力。因此，模组布局应与发热元件、散热风扇或散热孔保持一定距离。

## 9. Hardware Integration Considerations / 硬件集成注意事项

- 1) Use the VCC pin to provide a reliable power supply and all GND pins of the module are grounded;
- 2) Connect the ANT\_IN signal to the antenna, pay attention to the 50Ω impedance matching of the line;
- 3) The module reset pin SYS\_RSTN is for reset. Please connect it correctly to ensure that the module can be reset reliably;
- 4) Special attention should be paid to:
  - ①. Place the power supply as close to the module as possible
  - ②. Widen the power supply lines or use split copper surface to transmit current;
  - ③. Power routing should avoid high-power and high-inductive devices, such as magnetic coils.
  - ④. To ensure good heat dissipation of the module, it is recommended that the grounding pad of the module be fully in contact with the PCB and that more ground holes be added to facilitate heat dissipation

- 5) The PHY of the network uses a voltage type PHY, and when the external network port is connected, the middle tap of the transformer cannot supply power;
  - 6) Antenna interface: The antenna line is as short and smooth as possible, avoiding acute angles;
  - 7) Avoid wiring directly under K902;
  - 8) The module is as far away from the high temperature airflow as possible.
- 1) 用VCC引脚提供可靠的电源且模块所有GND引脚接地;
  - 2) 连接ANT\_IN信号至天线,注意线路50Ω阻抗匹配;
  - 3) 模块复位引脚SYS\_RSTN为复位, 请正确连接以保证模块可以可靠复位;
  - 4) 在设计中应特别注意:
    - ①布局上供电尽量靠近模块放置;
    - ②加宽电源走线或采用分割铺铜面来传输电流;
    - ③电源走线避免经过大功率与高感抗器件, 如磁性线圈;
    - ④为保障模块良好散热, 建议模块接地焊盘与PCB充分接触, 多加地孔便于散热。
  - 5) 网络的PHY使用的是电压型的PHY, 在外接网口时, 变压器的中间抽头不能供电;
  - 6) 天线接口: 天线线路尽量短且顺畅, 避免走锐角;
  - 7) 避免在K902正下方走线;
  - 8) 模块尽量远离高温气流。

## 10. Static Electricity Protection / 静电保护

Some components on K902 module are easily damaged by static electricity, which in turn affects the IC circuit and other components. Therefore, you should pay attention to electrostatic protection measures when using it.

- 1) When handling the module, try to wear gloves or finger cots and an anti-static wrist strap that meets the electrostatic protection standards;
- 2) During the process of taking the module, only the edge of the board should be taken, and the solder joints, circuit parts or components should not be directly touched to avoid sweat fingerprints from contaminating the solder joints;
- 3) The module should be protected by a soft protective pad between the module and the module interval during transportation;
- 4) When the module is idle, it should be placed on a soft protective pad (such as anti-static sponge pad, etc.), and do not stack at will;
- 5) The modules should be placed neatly and orderly, with a certain interval between modules to avoid collision with each other;
- 6) The module should be handled with care during use to prevent the module from being damaged by rough operation;
- 7) When powering on, pay attention to the positive and negative poles of the power supply and the voltage to avoid reverse connection and excessive voltage from burning the module;
- 8) When soldering the module to the motherboard, please ensure that the GND is soldered first, and then the ANT\_IN pin;
- 9) When handling ANT\_IN pin, do not touch any live capacitance or material (e.g., surface mount antenna, coaxial cable, wire iron, etc.) to avoid damage to the ANT\_IN pin by the charge generated or stored by said capacitor or material;



10) Please make sure to solder the ANT\_IN pin with an electrical protection iron.

K902模块上的部分元器件易受静电影响而损坏，进而影响IC电路及其他元件。因此在使用时应注意做好静电保护措施。

- 1) 拿取模块时应尽量戴好手套或者指套以及符合静电防护标准的防静电腕带；
- 2) 模块拿取过程中应只拿取板卡的边缘部位，不能直接接触焊点，线路部分或者元器件，避免汗液指印污染焊点；
- 3) 模块在运输过程中模块与模块间隔之间应该采用软性防护垫进行保护；
- 4) 模块闲置时应放置在软性防护垫上（如防静电海绵垫等），不要随意堆叠；
- 5) 模块摆放应摆放整齐有序，模块之间保持一定间隔，避免相互碰撞；
- 6) 模块在使用过程中应该轻拿轻放，防止粗暴作业损坏模块；
- 7) 上电时，注意电源正负极以及电压，避免反接和电压过高烧毁模块；
- 8) 将模块焊接到主板时，请确保GND先焊接，然后再焊接ANT\_IN引脚；
- 9) 处理ANT\_IN引脚时，请不要接触任何带电电容或材料（例如表贴天线、同轴电缆、电烙铁等），以免所述电容或材料所产生或存储的电荷损坏ANT\_IN引脚；
- 10) 请确保使用带电保护的电烙铁焊接ANT\_IN引脚。



## 4. Assembling & Repairing Note / 装配及维修说明

### 4.1. Module Assembling Note / 模块装配说明

K902 is surface mounted, SMT welding is used for assembly.

K902为表贴式模块，使用SMT的焊接方式进行装配。

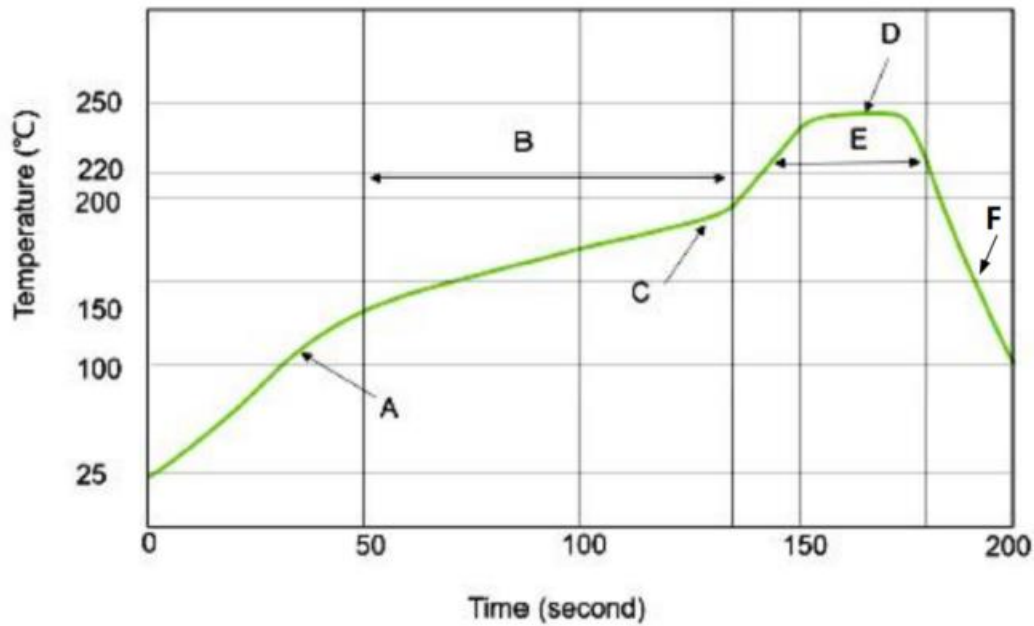


Figure 4. Furnace Temperature Curve / 炉温曲线

The process temperature limits are as follows:

- A: Heating Zone: Rising Slope:  $1 \sim 3^{\circ}\text{C} / \text{sec}$
- B: Constant Temperature Zone: Range:  $150 \sim 190^{\circ}\text{C}$  Time:  $80 \sim 110\text{ S}$
- C: Constant Temperature→Reflow Zone: Rising Slope:  $1 \sim 3^{\circ}\text{C} / \text{sec}$
- D: Peak Temperature:  $235 \sim 245^{\circ}\text{C}$
- E: Reflow Zone: Range: Over  $220^{\circ}\text{C}$  Time:  $50 \sim 80\text{ S}$
- F: Descent Slope:  $-5^{\circ}\text{C} \sim -1^{\circ}\text{C} / \text{sec}$

制程温度界限如下：

- A: 升温区：斜率： $1 \sim 3^{\circ}\text{C} / \text{sec}$
- B: 恒温区： $150 \sim 190^{\circ}\text{C}$  时间： $80 \sim 110\text{s}$
- C: 恒温→回流区：斜率： $1 \sim 3^{\circ}\text{C} / \text{sec}$
- D: 峰值温度： $235 \sim 245^{\circ}\text{C}$
- E: 回流区：大于 $220^{\circ}\text{C}$  时间： $50 \sim 80\text{s}$
- F: 下降斜率： $-5 \sim -1^{\circ}\text{C} / \text{sec}$

In order to prevent the module from being damaged by repeated heating, it is recommended to place the module after finishing the first side of PCB board.

为避免模块因反复受热而损坏，建议在完成PCB板第一面的回流焊之后再贴模块。

#### 4.2. Repairing Note / 维修说明

When disassembling the module, it is suggested using a BGA welding bench. Please use correct air tuyere and choose certain temperature curve. Keep peak temperature under 245°C, rising slope under 3°C /s.

拆卸模块时，请使用BGA返修台，选择适合尺寸的风嘴并使用合适的温度曲线，最高温度不超过245°C，升温斜率不超过3°C/s。

## 5. Application Connection Example / 应用连接示例

### 5.1. Minimum Circuit Description / 最小电路说明

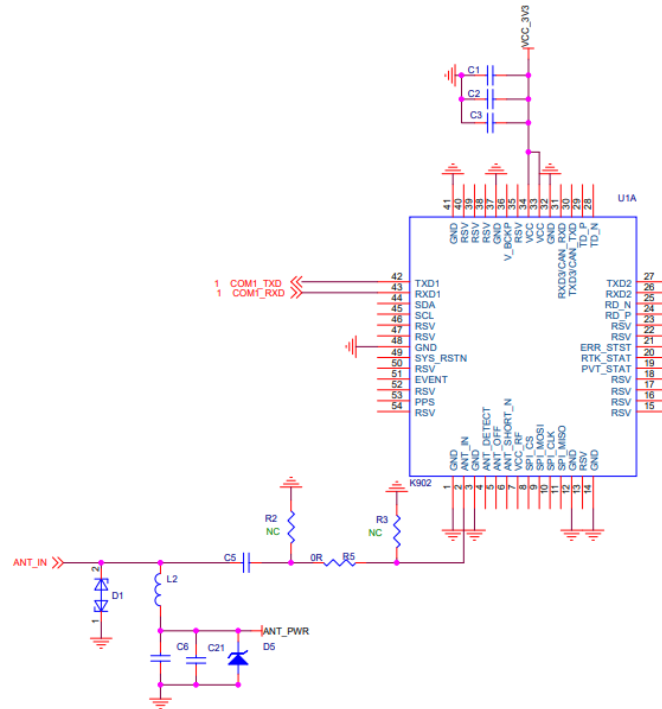


Figure 5. Minimum Hardware Design of K902 / K902 最小硬件设计

Table 5. Minimum Circuit Description of K902 / K902最小电路说明

| 位号    | 类型     | 数值    | 作用                                        |
|-------|--------|-------|-------------------------------------------|
| C1,C2 | 去耦电容   | 22uF  | 抑制低频波纹，稳压                                 |
| C3    | 去耦电容   | 1uF   | 抑制高频波纹，稳压                                 |
| C21   | 去耦电容   | 0.1uF | 抑制高频波纹，稳压                                 |
| C6    | 去耦电容   | 1uF   | 抑制低频波纹，稳压                                 |
| L2    | 馈电电感   | 68nH  | 射频阻断，推荐使用68nH射频电感                         |
| D1    | ESD二极管 | -     | 防静电，支持高频信号（2000MHz以上）的ESD防护器件             |
| D5    | TVS二极管 | -     | 避免出现过压导致元件损坏，需根据馈电电压、天线耐压等指标选择钳位特性达标的TVS管 |
| C5    | 隔直电容   | 100pF | 隔直，防止电压倒灌模块                               |

|       |    |            |                                 |
|-------|----|------------|---------------------------------|
| R2,R3 | 电阻 | -          | 预留 $\pi$ 型电路，确保50 $\Omega$ 阻抗匹配 |
| R5    | 电阻 | 0 $\Omega$ | 预留器件位置，保证匹配                     |

## 5.2. Transformer Connection Mode / 变压器连接方式

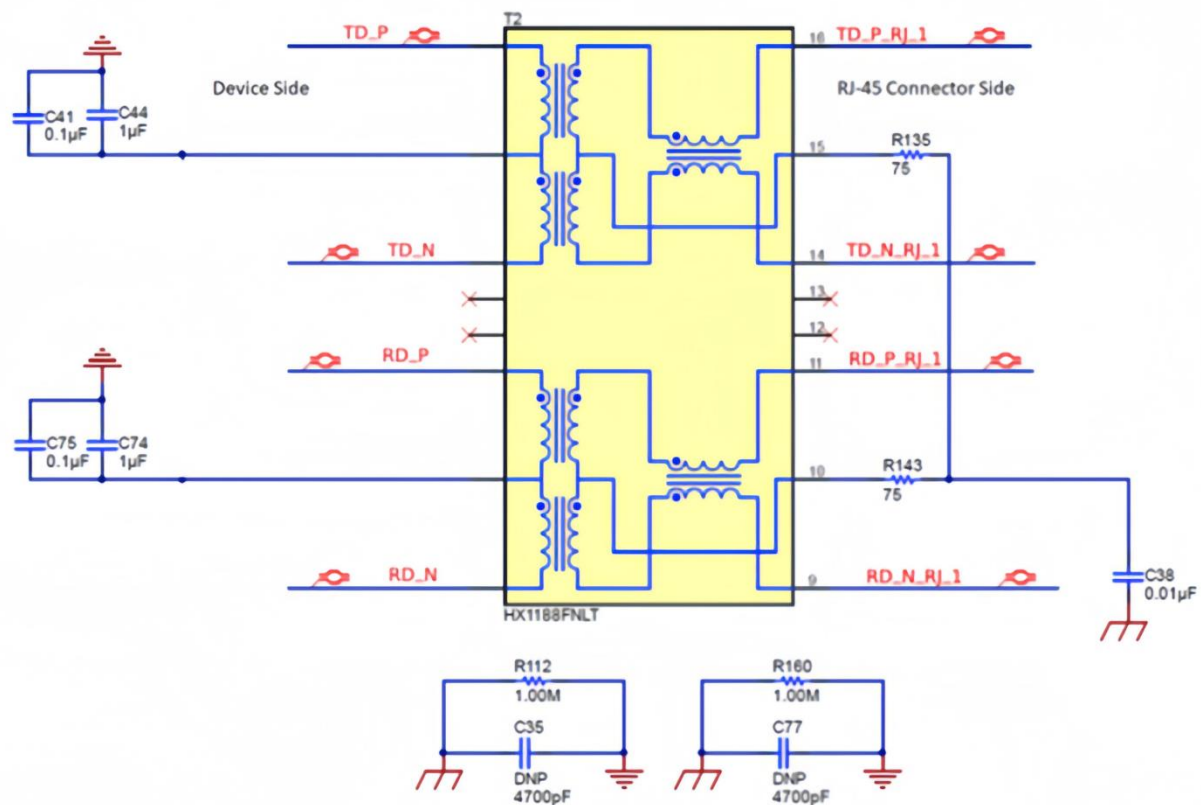


Figure 6. Transformer Connection Mode of K902 / K902 接变压器方式

Notice:

- 1) K902 Ethernet applications need a transformer;
- 2) The PHY of the network uses a voltage type PHY, and when the external network port is connected the middle tap of the transformer cannot supply power.

注意:

- 1) K902 以太网应用需接变压器;
- 2) 网络的 PHY 使用的是电压型的 PHY，在外接口时，变压器的中间抽头不能供电

## 6. Package / 包装

K902 module is packaged by means of carrier tape and coil (applicable to mainstream surface mount equipment), packaged in a vacuum-sealed aluminum foil electrostatic bag (The storage life of the K902 module in this anti-static bag is one year), containing desiccant for moisture proof. When the module is transported or not used after unpacking, it is still necessary to do moisture-proof work to avoid module failure. When welding modules by reflow soldering process, please strictly comply with IPC standards for humidity control of modules. Because the packing materials such as the carrier belt can only withstand 55°C, the modules need to be removed from the packaging during baking operation.

K902 模块使用载带、卷盘方式（适用于主流表面贴装设备），包装在真空密封的铝箔防静电袋中（K902 模块在此防静电袋中的保存期限为一年），内含干燥剂防潮。在模块运输过程或拆开包装后未使用的情况下，仍需要做好防潮工作，以免模块失效。采用回流焊工艺焊接模块时，请严格遵守 IPC 标准对模块进行湿度管控。由于载带等包装材料只能承受 55°C，在进行烘烤作业时需要将模块从包装中取出。

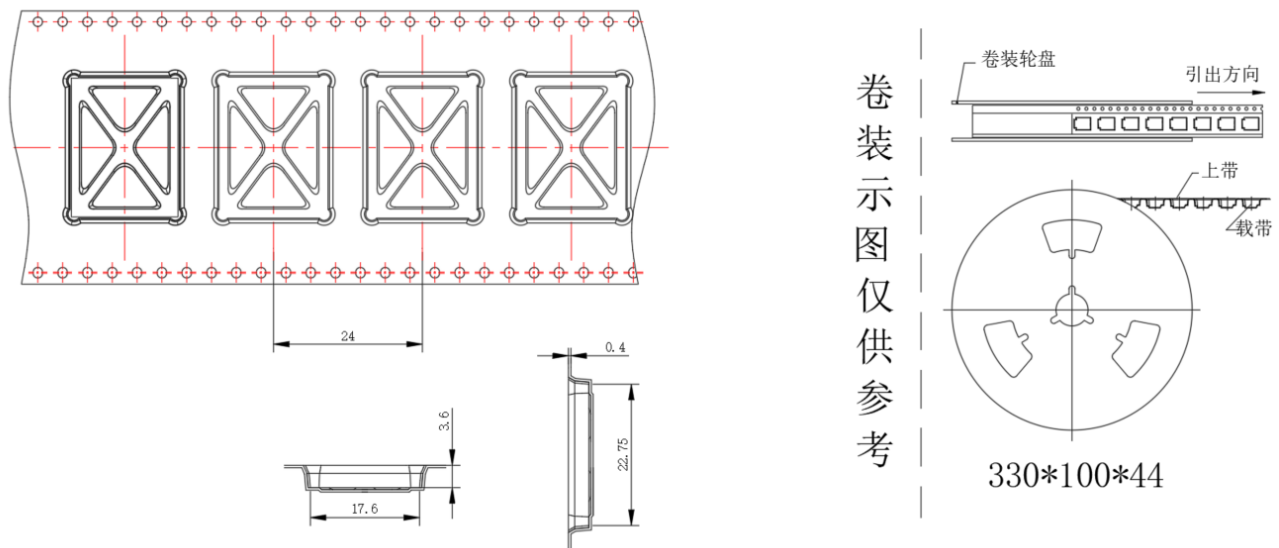


Figure 7. K902 Roll Tape Packing / K902 卷带包装

Table 6. K902 Package Description / K902 包装说明

| Project                  | Description                      |                                                                                                                                                                                                                                                                                                                                                         |
|--------------------------|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Module packaging<br>模块包装 | Number of Modules<br>模块数量        | 500 Slice/Roll<br>500 片/卷                                                                                                                                                                                                                                                                                                                               |
|                          | Reel Size<br>卷盘尺寸                | Material tray:13 inches<br>料盘: 13 寸                                                                                                                                                                                                                                                                                                                     |
|                          |                                  | Outer diameter: 330mm, inner diameter: 100mm, width: 44mm, wall thickness: 3.6mm<br>外径 330mm, 内径 100mm, 宽 44mm, 壁厚 3.6mm                                                                                                                                                                                                                                |
|                          |                                  | Package of each module: length 22.75mm, width 17.6mm<br>每个模块包装: 长 22.75mm, 宽 17.6mm                                                                                                                                                                                                                                                                     |
|                          | Carrier Belt<br>载带               | Module Spacing (Center Distance): 24mm<br>模块间距 (中心距) : 24mm                                                                                                                                                                                                                                                                                             |
|                          | Quantity of Desiccant<br>干燥剂数量   | 1 pcs/box<br>1 袋/盒                                                                                                                                                                                                                                                                                                                                      |
|                          | Number of Humidity card<br>湿度卡数量 | 1 piece/box<br>1 张/盒<br><br>Attention: Check the humidity card label in the package before the patch, and the 30% identifier ring of the humidity card should be displayed as blue if normal; If the color of the 20% label circle of the humidity card is pink and that of the 30% label circle is lilac, bake the card as required before mounting it |

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|                        |                             |                                                                                       |
|------------------------|-----------------------------|---------------------------------------------------------------------------------------|
|                        |                             | 注意：贴片前需查看包装内湿度卡标识，湿度卡的 30%标识圈颜色正常应显示为蓝色；若湿度卡的 20%标识圈颜色显示为粉色、30%标识圈显示为淡紫色，需按要求进行烘焙后再贴片 |
| Module storage<br>模块存储 | Storage Temperature<br>存储温度 | 23°C±5°C                                                                              |
|                        | Storage humidity<br>存储湿度    | ≤30%                                                                                  |

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